RESEARCH ROADMAP

MATCHED GNN-PPO / LSTM-PPO UPDATE - 2026-10-08

The revised manuscript now compares only GNN-PPO with the repository's matched LSTM-PPO node-sequence encoder. Five checkpoints per architecture were evaluated on the same 108 scenario IDs per checkpoint (540 episodes per method). The table reports simulator results with uncertainty across fitted checkpoints and shared scenario clusters; it is not field validation.

Measure	GNN-PPO	LSTM-PPO / paired result
Route completion (GNN-PPO)	249/540 (46.1%); nested 95% CI 31.3%-58.5%	Per-seed range 18.5%-63.9%; successful-hop checkpoint mean 87.77
Route completion (LSTM-PPO)	403/540 (74.6%); nested 95% CI 58.0%-87.6%	Per-seed range 42.6%-89.8%; successful-hop checkpoint mean 71.39
Paired completion effect		LSTM minus GNN +28.5%; nested 95% CI +20.6% to +37.2%
Matched outcomes		Both succeeded 242; GNN-only success 7; LSTM-only success 161; both failed 130
Successful-route hops		Among 242 episodes where both succeeded, mean LSTM-minus-GNN difference -32.56 hops
CPU actor-call p95 (GNN-PPO)	9.610 ms	Across-checkpoint SD 1.647 ms
CPU actor-call p95 (LSTM-PPO)	6.241 ms	Across-checkpoint SD 0.216 ms

Interpretation and remaining scope

The paired bootstrap is the primary uncertainty summary. Successful-hop differences are conditional on both methods completing the same scenario. The LSTM reads node-feature rows in a fixed topology order, has no graph-edge input, and resets its state on every action call; the comparison is therefore between these two implementations, not graph memory versus persistent temporal memory. Reward-component totals are descriptive because trajectories can differ in length. The manuscript figures are constrained to the sections that interpret them, and the publication bundle excludes older route-baseline and GNN/MLP comparisons retained in the local research archive.

The study is simulator-only. Field channel calibration, absolute key-pool and multi-demand service modeling, network blocking/capacity, broader topology tests, and a formal composable-security proof remain outside the completed paper revision.

The selected-seed reward audit is complete on 108 matched scenarios. Its cumulative reward-component report and episode ledger are included in the supplementary package; the plot is a descriptive diagnostic only.